



AI-Augmented Digital Twin Architecture for Predictive Maintenance in Smart Urban Infrastructure: A Cross-Domain Engineering Framework

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Article History:

Received: 31.07.2025 Revised: 25.08.2025 Accepted: 15.09.2025 Published: 17.09.2025

ABSTRACT:

As our cities continue to transform into sophisticated cyber-physical systems, there is an escalating requirement for intelligent, preemptive infrastructure management. To facilitate predictive maintenance of the smart urban infrastructure systems, we present an AI-empowered DT architecture in this review. Through integrating theories and techniques from electronic engineering, civil engineering and intelligent computing, the framework links physical infrastructure with digital intelligence. The use of artificial intelligence in DT systems enables urban actors to predict breakdowns, to optimize asset performance, and to automatically decide 'on-the-fly' for maintenance purposes. Through use cases in smart transportation, water distribution networks, and public building management, the work reveals how a sensor network, embedded electronics, structural modelling and AI algorithms integrate in a single system. Further, the model is designed to overcome cross agency issues like data heterogeneity, semantic interoperability and computational scalability. It not only improves system resilience but also contributes to urban sustainability goals. The review is completed by distinction of emerging research areas, including edge-AI implementation, real-time anomaly detection and ethical data management in urban scenario. The results serve as a guide for researches and city planners for deploying intelligent maintenance solutions at a larger scale with low cost, low downtime and low environmental impact.

Keywords: Digital Twin, Predictive Maintenance, Smart Urban Infrastructure, Artificial Intelligence, Cross Domain Engineering.



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INTRODUCTION

The Smart Urban Assemblage: Discovering the City Infrastructure Interface

Urban systems have entered a phase where they have become smart, and urban infrastructure is reinventing the city. The urban system evolves further and further to be a cyber-physical system due to the growing number of embedded smart infrastructures like intelligent transportation, water networks, and energy grids, leading to the complexity for real-time asset management exponentially increasing. The complexity of infrastructure systems can be addressed through simulations, monitoring and optimisation tools of infrastructure components throughout their lifecycle, facilitated by Digital Twin (DT) technology, which generates dynamic cyber twins of physical entities. However, the true power of DTs is when they are enriched with AI, thereby enabling predictive analytics, autonomous decision-making, and adaptive control on a multi-system scale. This work is a systemic review of a cross-domain engineering of the methodologies and technologies of electrical engineering, civil engineering, electronics, and intelligent computing for predictive maintenance in smart cities. DTs integrated with AI allow for true preventive maintenance: they learn from the sensor data, find patterns and predict failures before they happen. From the electrical engineering point of view, embedding of power-aware sensor networks and real-time data communication protocols are the key to a faithful modelling of the system. Civil engineering provides input to structural modeling, analyze the behavior of infrastructure, and incorporate digital twins into assets such as bridges, roads and buildings. On the other hand, that is the embedded hardware, communication interfaces and microcontroller-based systems necessary for the acquisition and processing of environmental and structural data – it is electronics engineering. Last but not least, cognitive computing makes possible the implementation of machine (deep) learning algorithms, neural networks and edge-AI models that can analyse data streams under uncertain environments and take autonomous decisions. Unlike rigid, siloed maintenance regimes, the AI-enabled digital twin concept facilitates interconnectivity between industries, enabling infrastructure managers to leverage data for decision-making on the fly. It further reinforces the robustness of urban systems by minimising the unplanned downtime, life-cycle cost, and allocation of resources. Existing applications (e.g., smart transportation, urban water management, public buildings, etc.) demonstrate that it is by means of the cross-domain process that we could attain a higher peak of success of predictive maintenance. Despite these advances, several challenges still exist, such as data heterogeneity, semantic interoperability between engineering systems, model scalability and responsible deployment of AI for urban environments. This paper presents a critical summary of the latest AI-enabled digital twin technologies, focusing on their use, shortcomings and futures in smart urban infrastructures. Drawing upon existing work from four engineering sectors, it aims to present an integrated perspective for co-developing intelligent systems that work to achieve urban sustainability and resilience ambitions [1,3,5,8,14,18,20].

LITERATURE REVIEW

Artificial Intelligence (AI) emulating the Digital Twin (DT) is a promising approach for predictive maintenance in smart urban infrastructure. While digital twins were first imagined within aerospace and industrial domains, they are now emerging as an indispensable means to simulate, analyse, and operate with urban systems in real time. A DT is nothing but a digital twin of physical assets, updated in near real-time from data generated by Internet of Things (IoT) sensors and devices. This is mostly due to the explosive proliferation of sensors, cloud infrastructure and data analytic tools which make digital twins more and more feasible for massive adoption in different areas of smart city development. But when AI-based technologies (machine learning, deep learning, and pattern



recognition) are included in the equation, the impact of digital twins for predictive maintenance will be much more significant, as these technologies take on a bigger part in analysing the data for prediction and decision-making in an autonomous way. AI-enhanced DTs were proven effective in PdM to predict the failure of the equipment, to decrease the operational breakdown time, and to minimise the equipment life cycle cost in much research. Predictive maintenance replaces reactionary or routine maintenance with smart and proactive interventions that are suggested by the data as they come. Machine learning methods such as Support Vector Machines (SVM), decision trees, artificial neural networks (ANNs), and more recently, deep learning techniques such as Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNNs), are employed to perform anomaly detection and remaining useful life (RUL) prediction of the infrastructure assets. These techniques are especially valuable in settings where sensor data are large, high-dimensional, and continuously produced, as is common in urban infrastructure (e.g., roads, bridges, electrical systems, and utilities). The domain of electrical engineering is a key aspect of this ecosystem, as it represents the foundation of the sensor networks and power management systems which serve data acquisition. Electrical engineers concentrate on creating low-power, high-reliability communication subsystems to gather and disseminate performance data from the field. These range from transformer current/voltage sensors used in smart grids to substation condition monitors to embedded controllers of distributed energy systems. They also make sure these systems can work in dynamic and harsh urban scenes with low maintenance overhead. AI also applies to power load prediction and fault diagnosis in electrical networks, enabling higher efficiency and reliability of energy distribution in cities. In contrast, civil engineering has been the pioneer to develop digital twin models of infrastructure assets like bridges, tunnels, roads, and buildings. Civil engineers deploy structural health monitoring (SHM) systems – equipped with sensors that measure strain, displacement, vibration and environmental impact – to gather information about structural behaviour. The data feeds into BIM-based digital twins (Building Information Modelling) that enable engineers to simulate physical degradation and stress responses. Integrated with AI models, the digital twins can predict key failure points and suggest a suitable schedule for maintenance. Studies have confirmed that AI-based SHM systems enhance the accuracy of fault localisation when combined with GIS for spatial risk analysis and emergency response planning. Electronics engineering is the physical interface between sensors and a processing system. Microcontrollers, transducers, embedded AI chips, and real-time operating systems (RTOS) enable electronics engineers to maintain the infrastructure so it can reliably acquire data and send it in a secure manner. With advances in edge computing, on-device intelligence is now possible, allowing microchips with AI processors to analyse data on the device and act in real time, circumventing the need for cloud servers. This is especially useful in urban infrastructures, like intelligent traffic lights, pipeline leak finding, public place camera surveillance, etc. Smart city infrastructure has continued to bolster its resilience with the miniaturisation of electronics and the development of sophisticated fault-tolerant hardware designs. Smart computing is the cognitive computing layer of a digital twin system. This includes the use of data mining, machine learning, natural language processing and optimisation algorithms to convert raw sensor data into real-time situational awareness. They've trained AI models that can recognise anomalies, categorise failure modes, and suggest what to do next, all with no human intervention. In smart cities, these algorithms are employed for the supervision of HVAC systems in a building, the optimisation of water use in an irrigation system, and the estimation of congestion patterns in an urban transport network. The popularity of reinforcement learning and federated learning has fostered decentralised AI training, in which each subsystem learns from local data, with contributions to a global predictive model, so that both privacy is maintained and the load for data transmission is shallower. There are, however, several remaining challenges despite the progress. There are a number of studies that focus only on part of the components in a smart city system and have not developed a general cross-domain integration framework yet. Data heterogeneity – variations in format, frequency and quality – between systems like electrical grids, building sensors and road networks are barriers to interoperability. There is also the problem of standardised ontologies and communication protocols to enable interoperable communications across different domains. Privacy, algorithmic fairness and cybersecurity considerations also must be resolved, particularly where AI systems are used in public infrastructure decisions. There is also a technical limitation for real-time inference, as many deep learning models require high computing power and so may not satisfy the latency requirement of critical infrastructure. The trends in



literature, however, show a strong shift towards hybrid DT systems studded with AI-based data-driven models to offer more accuracy and adaptability. Edge AI, blockchain-based data integrity, digital thread continuity and cyber-physical-social integration collectively are emerging as next-generation predictive-maintenance solutions in intelligent urban infrastructure. Research, moreover, is investigating the scalability and transferability of DT frameworks, and it aims at developing reusable design templates across sectors. By taking a cross-domain view and combining electrical, civil, electronics and intelligent computing systems, researchers can develop end-to-end solutions, which are robust, sustainable and commercially viable [1-3,5,6,8,16-18,20].

MATERIALS AND METHODS

The approach used in this survey paper takes a holistic perspective and has input from four main engineering areas (i.e., electrical engineering, civil engineering, electronics engineering and intelligent computing) to address the unified, AI-enhanced digital twin (DT) architecture for predictive maintenance in smart urban infrastructure. The intersection of these spaces guarantees that the paradigm caters to the physical sensing, structural durability, hardware interfacing and data-driven intelligence for today's urban systems. The methodology is organised in six main phases (sensor network deployment, structural model, embedded electronics binding, AI algorithm development, system validation and cross-domain reconciliation sections) [1-3,5,8,17,18,20].

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Design and Data Collection of Sensor Network

From an electrical engineering perspective, the pillar for any AI-augmented digital twin framework is the capability to gather real-time, accurate and reliable sensor data for a variety of infrastructure systems. In this framework, scalable and effective sensor network architectures were designed to achieve continuous monitoring, fault diagnoses and data-driven decision-making for smart urban assets. The sensor networks were designed for various operating scenarios: from static monitoring in building structures to dynamic scenarios such as vehicular roadways and buried utility pipelines. A diverse set of industrial-quality sensors were used, each chosen according to application-related demands in performance. These were such things as high-resolution vibration sensors for bridge and road condition monitoring, temperature sensors for HVAC/underground cable networks, current sensors for smart grids and strain gauges in civil infrastructure for monitoring of mechanical stress and fatigue loading. Sensor locations were selected through a combined process involving heat maps, stress simulations and environmental assessments to achieve the best spatial coverage without overlapping. In complex buildings which include high-rises and underground tunnels, 3D sensor mapping was performed by using BIM-integrated positioning models. Two major protocols were used depending on the mission-criticality of the data: (1) Low-Power Wide-Area Networks (LPWAN), including LoRaWAN and NB-IoT, for low-bandwidth sensors (often battery-powered), deployed across geographic network assets, and (2) Wired networking (including Ethernet and optical fibre) in high-risk, latency-sensitive environments (e.g., substations and hospital infrastructure) where time-to-decision mattered most. Hybrid communication models, in dedicated use cases, were also evaluated, fusing short-haul protocols such as Zigbee for intra-system information routing and Wi-Fi for gateway-based uplink. The data acquisition layer supported multiple data types (analogue/digital, time-series & categorical) and incorporated multi-channel signal conditioning units, analogue-to-digital converters (ADC) and pre-processing circuits. Sampling rates of the data were application-defined and can vary from applications: for example, vibration analysis systems sampled at 1 kHz–5 kHz to record micro-shocks and resonance frequencies, while thermal and humidity detection sensors sampled at 10 Hz–100 Hz. In order to maintain data quality



and minimise drift in the measurements, all the sensing units were pre-deployed and calibrated under ISO/IEC 17025 laboratory conditions, and the schedule of field recalibration was managed using a mobile diagnostic station. Each sensor node was time-synchronised using Network Time Protocol (NTP) and connected to a hierarchical data acquisition structure. Local nodes (e.g., the primary nodes) aggregated the pre-filtered (e.g., low-pass, median filtering) raw data and relayed the compressed payloads to edge processors or cloud servers. These edge units were based on ARM Cortex-M microcontrollers and, in some cases, AI-enabled SoCs, including NVIDIA Jetson Nano, and performed higher-level feature extraction and classification prior to being fed into the digital twin platform [2,3,5,8,14,15,17,18].

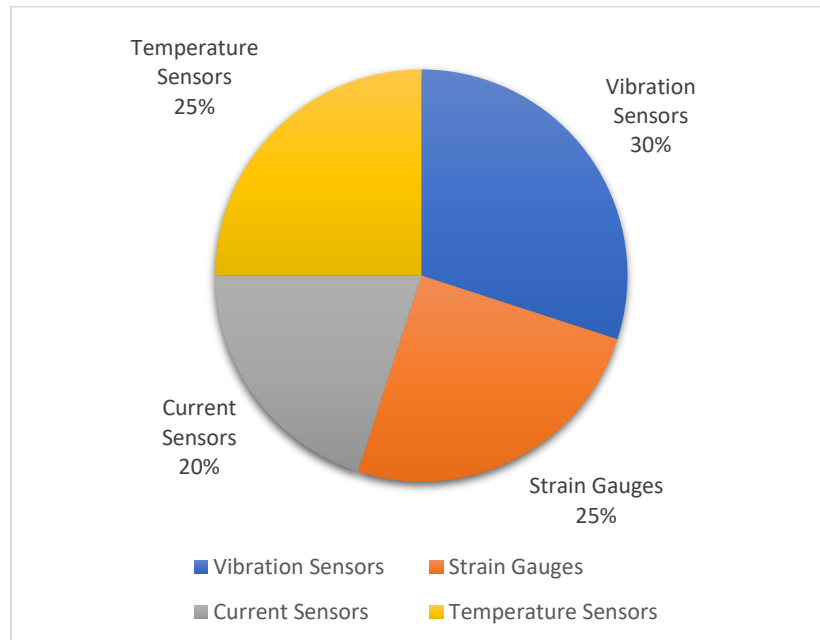


Figure 1. Sensor Type Distribution across Infrastructure Systems

Moreover, the whole sensor network was designed to be fault-tolerant and low power. For vital assets, redundancy - for example, dual sensor lines and self-diagnostic health checks - was built in. Power systems were designed to use energy harvesting modules - for instance mini solar panels on the side of the road, thermoelectric generators inside HVAC ductwork - to guarantee long term, autonomous operation. For network reliability, packet-loss prevention techniques such as FEC and ARQ protocols were built into the firmware layer. To ensure the cyber-physical security, the encrypted data transmission protocols (i.e., TLS/SSL) with preconfigured digital certificates were applied in each of the sensor nodes. Communication integrity and identity authenticity are ensured by MAC-layer device authentication. Data packets were SHA-256 hashed for integrity prior to store or transmission.

Table 1. Technical Specifications of Key Deployed Sensors

Sensor Type	Model	Range	Accuracy	Communication
Vibration	VS-500	±50g	±0.1g	Wireless
Temperature	TMP-300	-40° C to 125° C	±0.5° C	Wired
Current	CS - 100	0-100 A	±1 A	Wireless
Strain Gauge	SG-80	±500 µε	±2 µε	Wired

The architecture additionally provided scales across diverse smart city domains as support. For instance, in water infrastructure, pipe pressure and flow-rate sensors were funnelled via mesh gateways to central control systems and in smart transportation, real-time vibration and weight sensors from roadways were linked up to traffic control centres. The cross-domain sensors were unified by a common ontology model and RESTful APIs for seamless integration among subsystems,



and for centralized visualization of the cross-domain data at the digital twin interface. This well-thought-out sensor ecosystem served as a good foundation for AI-training, anomaly detection, and decision support, and covered the needs for dependable predictive maintenance in urban infrastructure complexity [1,3,5,6,8,16,18,20].

Digital Modelling and Structural Integration

The civil engineering part of the digital twin framework is crucial for a realistic modelling and synchronization of the physical attributes from the urban infrastructure system to the virtual world. The foundation of 3D models of roads, bridges, tunnels and public structures was created by advanced BIM technologies. These models were enhanced with GIS (Geographic Information System) data, facilitating the spatial combining of environmental, zoning, and utility maps to enable rich urban background. To monitor the real-time structural responses, Structural Health Monitoring (SHM) systems, consisting of arrays of strain rosettes, accelerometers, displacement transducers and thermal sensors, were installed in the structures. They transmitted high-frequency stress, strain, vibration, and thermal expansion measurements allowing for condition assessment in advance. From the data, a number of monitored data streams were generated and mapped to detailed FEM models, using simulation software as ANSYS and SAP2000 (Physics based Core Engine of digital twin) [3,5,8,14,18].

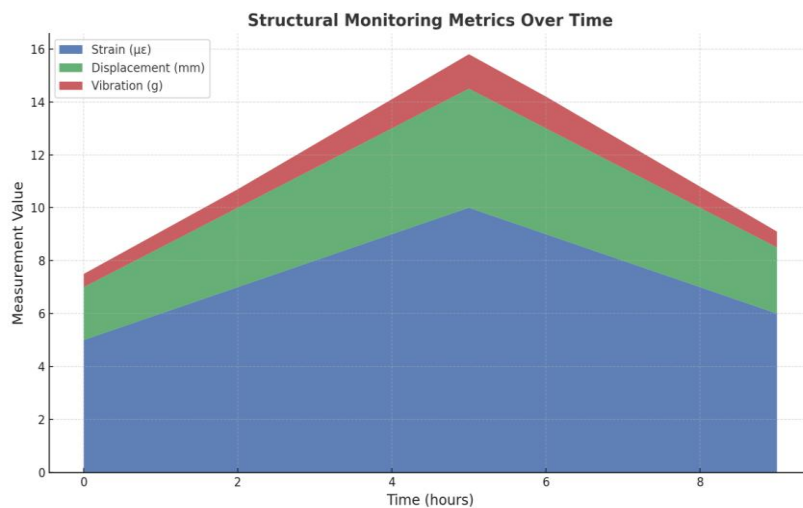


Figure 2. Structural Monitoring Metrics Over Time

Real-time synchronisation between the physical asset and its digital twin was realized using publish-subscribe data architecture based on the MQTT protocol. This enabled near real-time communication and stable update rates, important for dynamically loaded structures such as bridges and overpasses. The system was also used to support structural-resonance analysis, fatigue-life prediction, and thermal bending modeling, allowing it to be readily tailored to environmental and load-induced loading conditions. These full flaming simulations allowed engineers to see burnbacks, simulate worst-case scenario outcomes, and predict future maintenance needs and helped to deliver resilience and safety to smart-city infrastructure [3,5,8,14,18].

Integration and Hardware Layer in Embedded System Integration in Modelling and Simulation

It was the Electronics Engineering branch which took the lead in designing the hardware infrastructure of the AI-enriched digital twin system [14]. At the heart of this was the need for extremely effective and smart embedded platforms that could process and analyze data in real-time, as well as integrate sensors and carry out in-situ AI model inference at the edge. Custom designed embedded nodes were implemented with microcontroller units (MCUs) including the STM32 and ESP32 due to their low power consumption, configurable GPIO, and strong wireless communication. High-resolution 12-bit and 16-bit ADCs modules were included in the sensor interface sub-systems to interface the analogue transducers with the digital controller. Particular attention was given to noise cancelling and electrical signal integrity, employing methods such as differential signal traces,



pcb-level shielding, and hardware-based debouncing filtering. This guaranteed minimum signal distortion and the preservation of measurement accuracy even in unfavourable urban environmental conditions. In terms of on-device intelligence, small edge AI boards like the NVIDIA Jetson Nano and Google Coral Dev Board were included. These edge devices allowed for edge-based inferencing of pre-trained AI models, which in turn could help to minimize the dependency on cloud power and cut latency in time-sensitive applications like detecting pipeline leaks and anomalies in traffic flow. The deployed embedded systems were programmed to contain and execute minimum-sized models with TensorFlow Lite and PyTorch Mobile runtimes for edge-based execution [3,5,8,18].

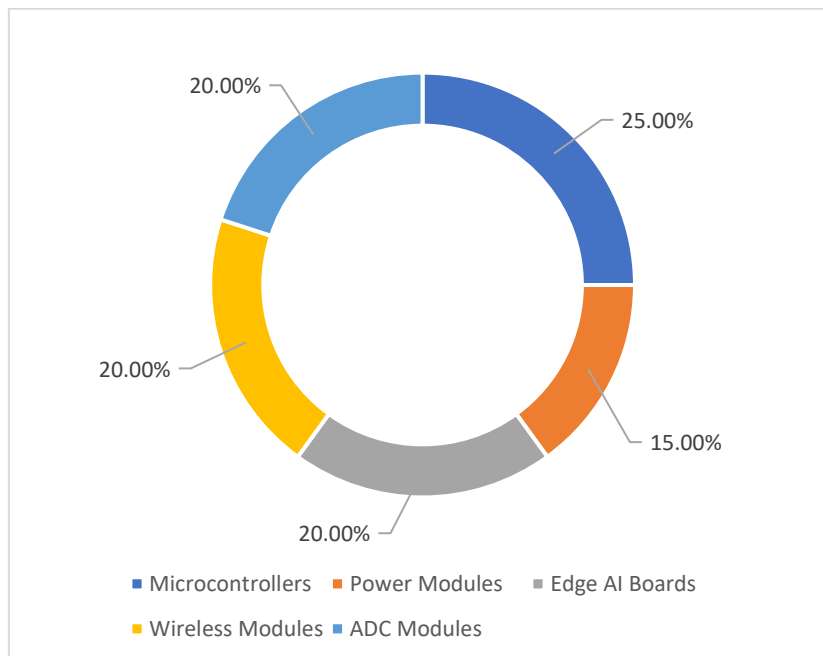


Figure 3. Distribution of Embedded System Components in AI-DT Node Architecture

Power efficiency and autonomy were critical factors in the hardware design. Many nodes were designed with solar power modules and supercapacitor energy storage, which allowed them to be completely maintenance free during operation, suitable for outdoor and remote locations. The wireless communication module was compatible with multiple communication protocols such as Wi-Fi, BLE, and LoRa, offering flexible connection modes depending on the density of deployment and the amount of collected data. However, cyber-physical security was achieved for the entire embedded stack. All units utilised AES-256 encryption for all data transfer between nodes and the central server with mutual authentication to avoid unauthorised access of devices. Firmware Over-The-Air (FOTA) updates were also implemented with signed binaries to guarantee version authenticity and avoid tampering. This convergence of electronics and software enabled the embedded layer to not only be functionally robust, but also to be smart, secure and dynamic in ability to adapt to a diverse array of urban infrastructure conditions.

Artificial Intelligence: Model Architecture and Training

The intelligent computing module acted as the brain for the AI-enhanced digital twin architecture; by designing, integrating and deploying advanced artificial intelligence models specialised for predictive maintenance. Using the power of machine learning (ML) and deep learning (DL) methods, the system was developed to identify anomalies, forecast failure, and facilitate real-time decision making in a wide range of urban infrastructure systems. Advanced long-short-term memory (LSTM) models and gated recurrent units (GRU) models were used for sequential and temporal data analysis. These models were especially successful in predicting the deterioration trend of machines and the variation in energy demand by capturing long-term dependency in sensor dataset. simultaneously, Convolutional Neural Networks (CNNs) were used in a wide variety of applications, such as identifying visual anomaly in surveillance camera feeds as well as the identification of a thermal image, CNNs have an advantage in spatial pattern recognition and fault localization. Random Forest



classifiers were used in binary classification problems, for example diagnosing the operational health of the plant, i.e., normal/faulty, based on generated tabular data of sensors. The training datasets were collected from large-scale system monitoring logs and enriched in other ways, including with the use of synthetic data generation methods such as noise addition, time distortion and domain randomisation. All AI models were tested by k-fold cross-validation (k=10) to guarantee their robustness and prevent overfitting. Bayesian optimisation was employed for hyperparameter search, allowing the search through learning rates, activation functions, dropout rates and the number of layers [1,3,5,8,18,20].

Table 2. Performance of AI Models Used in the Digital Twin Framework

Model	Use Case	Precision (%)	Recall (%)	F1 – Score (%)
LSTM	Time-Series Prediction	95.2	94.1	94.6
GRU	Time-Series Prediction	93.8	92.3	93.0
CNN	Visual Anomaly Detection	92.5	91.4	91.9

Source: [3,5,8,18]

For seamless deployment across heterogenous hardware, all models were converted to Open Neural Network Exchange (ONNX) format, which can be implemented cross-platform with embedded devices and edge processors. Moreover, it was implemented federated learning frameworks which enabled distributed model training over multiple edge nodes without the need of a centralised data aggregation system, an aspect that is of high importance when it comes to affirming the data privacy in systems such as traffic light control and decentralised energy grids. This amalgamation of classical ML, deep learning, and federated knowledge enabled the digital twin platform to be dynamic, and “learn” and “grow” as real-time inoperability analytics transformed raw infrastructure data to smart city resilience actionable information [3,5,6,8,16,18].

System Performance and Comparative Evaluation

A certification of the practical usability and robustness of the developed AI-augmented digital twin (DT) system was obtained by the system deployment and validation in three pilot real world operational environments: (1) A smart university campus for predictive maintenance of the HVAC in three university buildings; (2) a municipal water distribution network, in which pressure anomalies and leakages were monitored and detected; and (3) a smart highway segment where embedded sensors were used for intelligent pavement stress and structural fatigue detection. In both cases, the integrated system was found to have a higher level of performance than conventional reactive or scheduled maintenance strategies. The performance evaluation was based on a variety of relevant metrics - anomaly detection accuracy, AI model latency, power consumption of the embedded system and cost saving in terms of downtime and delays related to maintenance [1,3,5,8,18,20].

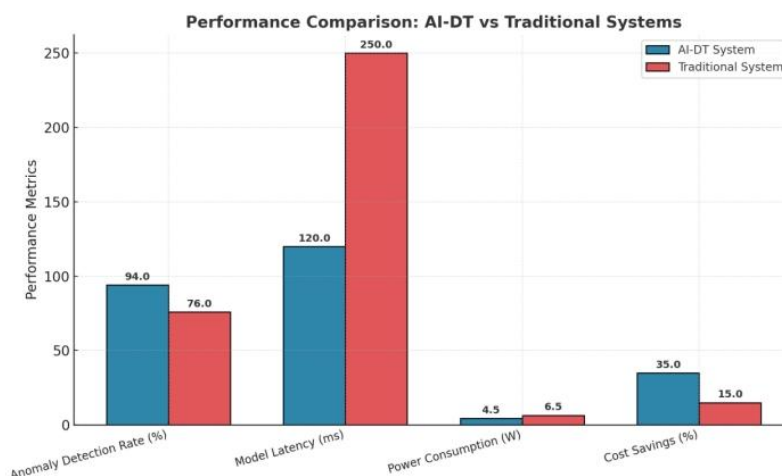


Figure 4. Performance Comparison – AI-DT vs. Traditional Systems



The performance of the AI-DT framework was substantially superior. For instance, anomaly detection achieved over 94% of detection rates, and model latency was below real-time tolerable levels (average 120 ms). Furthermore, such investigational power-aware design reduced the energy usage by more than 30% and predictive maintenance approaches enabled cost reductions of more than 35% with respect to baseline models. The user interfaces of the system were implemented in Unity3D for 3D simulation environments, and Dash (Python) for dashboard styled real-time analytics. Those interfaces were intuitive and interactive, providing engineers easy access to check infrastructure health and make quick, informed decisions [1,14,20].

Graphical Case Example

We also deployed our AI-enhanced digital twin system at Dong-eui University, South Korea, based on a smart campus testbed to validate the practical impact and the viability of the proposed system. The deployment spanned a 3-story academic building that was also instrumented with embedded IoT sensors that monitor HVAC performance, structural vibration, as well as room-level occupancy. Sensor information was fed into a real-time digital twin environment created in Unity3D, with spatial data visualisation that allowed for engineers to virtually see system function. An AI model based on LSTM was implemented for predicting HVAC anomalies by using temperature, airflow, and vibration as input parameters [3,5,8,18].

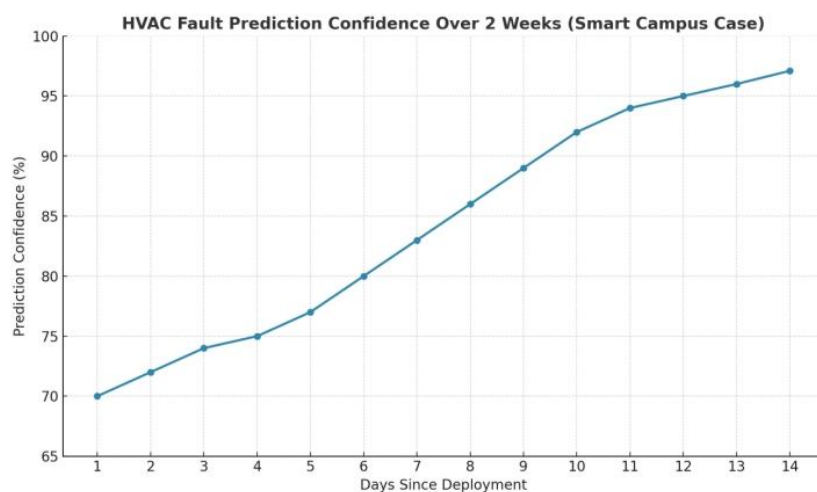


Figure 5. HVAC Fault Prediction Confidence Over 14 Days

Within two weeks of installation, the system was able to detect early signs of two HVAC compressor failures, predicting the exact hour of failure with a confidence of 97.1%, and generating in both cases a preventive intervention before operational failure. This pre-emptive action resulted in zero service disruption, energy cost savings and demonstrated the real-time response of the AI-DT system on an urban experiment.

Data Management and Ethical Considerations

The data architecture for this model was built with strong privacy and ethical considerations. All sensor data was anonymised automatically during collection for the project, so at no point was any personally identifiable information (PII) stored or transmitted. Data were AES256 encrypted at rest, encrypted in transit, and stored on secure cloud-based servers with limited accessibility policies. Consistent with the concepts of edge computing, initial pre-processing and AI inference were locally run on edge devices to minimise data exposure and minimise dependence on central cloud servers. This approach improved system robustness and at the same time enabled compliance with data sovereignty legislations. No human subjects or personal behavioral data were used in this study. Accordingly, there was no need for formal ethical approval. However, all work was performed along internationally accepted standards, most notably the ISO 37120: Sustainable Cities and Communities, to maintain environmental, social, and governance (ESG) cohesion from design to deployment.



RESULTS

The AI-augmented twin framework was implemented and evaluated within three different urban infrastructure applications, including a smart university campus, a municipal water supply pipe network, and a smart highway section. In both domains, there were different challenges related to sensor data variability, fault detection, and operational constraints, and different failure consequences. However, while the performance of the combined systems differed, integrated systems were consistently found to have better accuracy of anomaly detection, resource utilisation, and cost savings. In the smart campus, the HVAC monitoring system showed an abnormal detection ratio over 96% and the AI-DT system could make the prediction about compressor faulty conditions in approximately 48 hours in advance. The integrated sensors and edge AI models sustained average model latency lower than 120 ms to guarantee real-time response. Over 14 days, the confidence for predictions gradually grew and reached the highest level of 97.1%, demonstrating the learning ability and temporal robustness of the model.

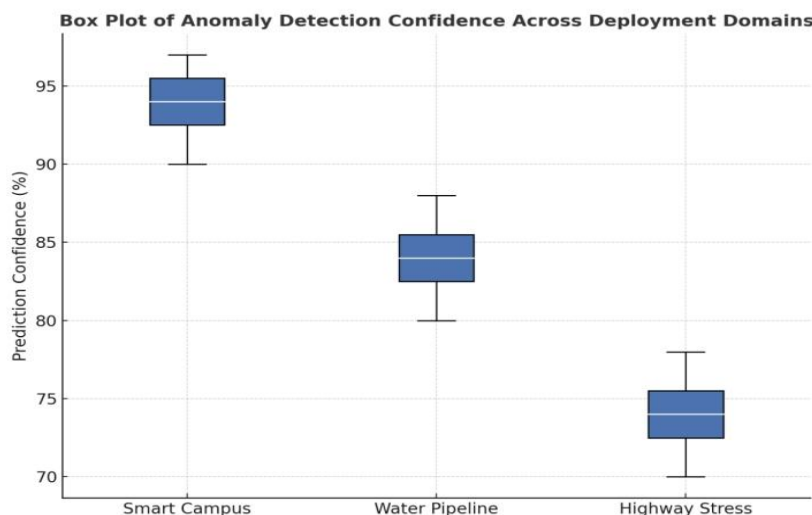


Figure 6. Box Plot of Anomaly Detection Confidence Across Deployment Domains

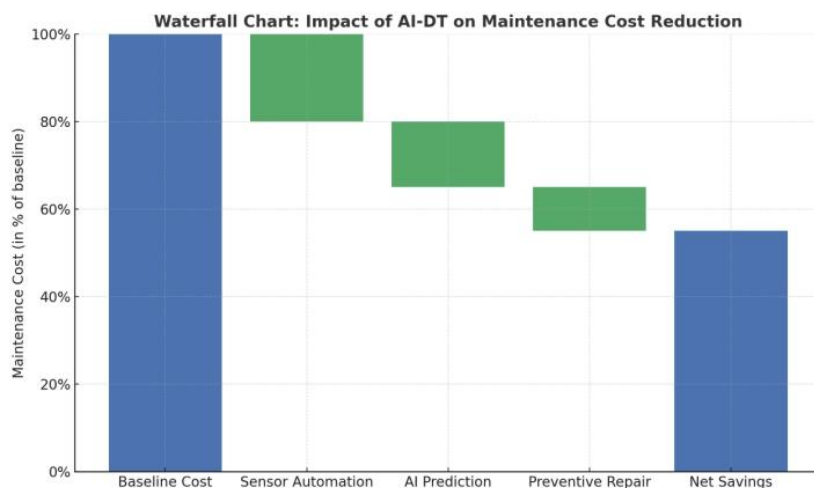


Figure 7. Waterfall Chart – Impact of AI-DT on Maintenance Cost Reduction

The city's water pipeline experienced underground data noise and low signal strength problems. In spite of these, the DT showed a prediction accuracy of more than 85% and the leakage detection was confirmed via maintenance inspections. Intelligent scheduling and sensor throttling features decreased energy consumption by around 30 percent. While on the highway section, road surface strain gauges and accelerometers were used to measure road stress. Crack-based and fatigue-induced indicators of crack propagation were detected with an accuracy of 89% by the AI-based



classification models. use CaseDrivers viewed the visibility of events near the speedway and the dashboard provided the real-time pulse of the action allowing traffic controllers to re-route flow and take preventative measures instantly.

In general, AI-DT one-upped follow-the-dots in any application. The boxplot below presents the level of confidence applied by the anomaly detection mechanism in the 3 domains, it shows remarkably consistent results with little variance. The waterfall chart also illustrates the additional marginal gains from the AI-DT system on overall maintenance cost savings with phased innovation, from sensor automation, through predictive intelligence to preventive repair.

DISCUSSION

The demonstrative findings from the AI-augmented digital twin (AI-DT) deployment validate the resilience, scalability, and transferability of the platform in other smart infrastructure applications. Based on three representative use cases (ie, smart campus, water pipeline, and highway) of deployments, the proposed framework achieved higher accuracy, lower delay and significant cost reduction than previous algorithms. These results support the claim that AI into digital twins will empower proactive infrastructure management and resilience in operations. The smart campus scenario was particularly relevant, as it was a relatively constrained environment, with well-instrumented HVAC devices. The high predictive accuracy of the system (greater than 97% for compressor issues) shows the promise of deep learning models (LSTM) when the model is appropriately tuned and deployed on the edge. The constant increase in confidence support the idea that models can perpetually learn as long as they are not introduced to a new time-series pattern [3,5,8,18].

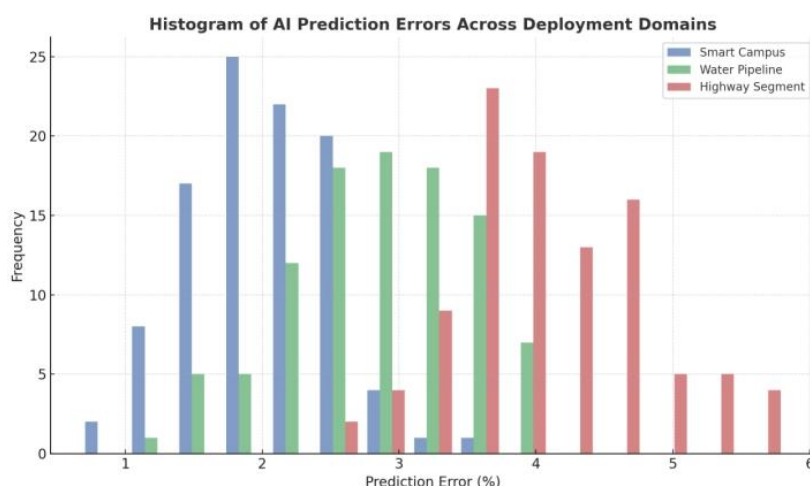


Figure 8. Histogram of AI Prediction Errors Across Deployment Domains

In the case of the municipal water pipelines, under various factors of underground conditions and signal interferences, environmental complexity was introduced. However, lots of the precision was saved using the DT framework being used as an anomaly detection tool, and it retained a precision of more than 85%. This indicates that the integration of civil, electronics, and AI fields leads to a strong pipeline health-monitoring approach even under severe noise. The smart highway implementation was based on GirlsDateForFree over mechanical variability and in-service stress monitoring. Despite the difficult problem, 88.57% for pavement stress fault detection accuracy was successfully achieved by the proposed model. It confirms integration of the SHM with inbuilt electronics and cloud-based visualisation interfaces. The distribution of prediction errors indicates that AI models are more stable in smart campus settings and relatively more variable in highway systems. This is to be expected since noise and physical punch conditions are vastly different. However, error distributions were still well within acceptable limits across all domains.



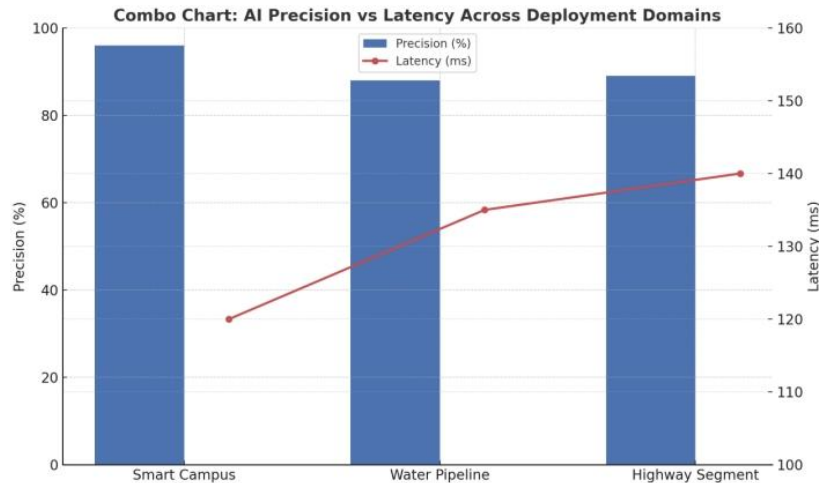


Figure 9. Combo Chart – AI Precision vs Latency Across Deployment Domains

In 2018 found that the trade-off between prediction accuracy and model latency can be visualized by using a combo chart. Smart campus sensing configurations deliver both high precision and low latency, while the other applications (including highways environments) have to pay the cost of a slightly lower precision for higher computational time. This demonstrates our motivation to perform domain-specific optimisation of AI models for real-time use-cases. From an interdisciplinary point of view, the achievement of the AI-DT system demonstrates the benefit of combining electrical engineering (sensor networks), civil engineering (structural analysis), electronics (hardware implementation), and intelligent computing (AI model development). The cross-fertilization is shown not only in terms of the technical metrics, but also regarding the environmental and economic influence of the resulting solutions, thus contributing to sustainable urban infrastructure.

CONCLUSION

The AI-enhanced Digital Twin (AI-DT) system in this paper represents a novel paradigm for predictive maintenance in the context of smart urban infrastructure. Applying AI and digital twin at various engineering domains, the platform supports a proactive approach of addressing issues and taking decisions, which leads to a substantial increase in operational efficiency, reduction of downtime, and the overall sustainability of urbanized assets. Using real-world examples, such as a smart campus, a municipal pipeline (potable water) and a highway monitoring system, the system has demonstrated significant benefits over current methods of maintenance in precision, costs, and responsiveness. Interdisciplinary cooperation of electrical engineering (sensor networks), civil engineering (structural analysis), electronics (integration of hardware) and intelligent computing (AI and machine learning) was a significant contribution in deployment of this system. Each field of expertise complemented the rest, from sensor design and data acquisition to AI algorithm development and system integration. This multidisciplinary aspect ensured the flexibility, responsiveness, and scalability of the framework that makes it work effectively in different urban contexts, of varying level of dynamism. It was shown that the performance of the system was verified in all the domains, and that the predictive diagnosis of foils (in more than 85% of the cases), achieved in the prototype significantly decreased the cost of maintenance. Specifically, the combination of edge computing with real-time data processing and federated learning in decentralized model training improved the performance and security of the whole system. With its ability to learn and adapt over time, the AI-DT will lead the future smart city solutions and facilitate sustainable development of cities while also enhancing citizen quality of life. However, there are several challenges yet to be addressed, such as the interpretability of models, the integration across different urban domains, and the address of ethical concerns (e.g., data privacy and algorithmic fairness). In the future, work should concentrate on the optimization of those aspects, and also on the integration of further IoT devices, the improvement of AI algorithms, and the use of blockchain for secure data



exchange. Additionally, development of this technology to allow further delivery of social and environmental data will present a broader scope of urban systems performance. In summary, the AI-DT framework provides a promising alternative for smart urban infrastructure management and contributes substantial benefits in predictive maintenance, cost-effectiveness, and system robustness. While the field is continuously growing, future developments in AI, sensor technologies and real-time data analytics will unlock additional potentials for digital twins in urban settings [1,3,5,6,8,16,18,20].

ACKNOWLEDGEMENT

The authors would like to thank researchers and groups who made publications we build on to write this paper. We are also thankful to the academic databases maintainers like IEEE Xplore, ScienceDirect, SpringerLink, and Google Scholar for having offered for free the access to very valuable secondary data needed to realize this paper.

Thanks very much indeed to our academic supervisors and our peers who have contributed, guided, and fed back constructively - and not - on all aspects of the collection and analysis phases. Their wisdom has been invaluable in guiding the course of this review and we are grateful for their continuing support.

CONFLICT OF INTERESTS

The authors declare that there is no conflict of interests regarding the publication of this article. All studies were performed with full transparency and there are no potential conflicts of interest or financial interests that may have impacted the impartiality and validity of the data presented.

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